advanced data format session 2

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setting a working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraris

```
library(dplyr)
library(tidyverse)
library(ggplot2)
```

import data

```
gss <- read.csv("gsssubset.csv")
head(gss)
```

```
  id    sex         degree   income  marital age height weight hrswrk
1  1   MALE       BACHELOR 60967.50 DIVORCED  53     72    190     60
2  2 FEMALE       BACHELOR 60967.50  MARRIED  26     60     97     40
3  4 FEMALE       BACHELOR 10161.25  MARRIED  56     68    160     20
4 14 FEMALE    HIGH SCHOOL 17551.25  MARRIED  40     65    156     37
5 16   MALE    HIGH SCHOOL 17551.25  MARRIED  56     NA    210      6
6 19   MALE LT HIGH SCHOOL 15703.75  MARRIED  51     68    170     NA
```

double filter filtering

single filter

```
single_filtered <- gss %>%
    filter(marital == "MARRIED")
head(single_filtered)
```

```
  id    sex         degree   income marital age height weight hrswrk
1  2 FEMALE       BACHELOR 60967.50 MARRIED  26     60     97     40
2  4 FEMALE       BACHELOR 10161.25 MARRIED  56     68    160     20
3 14 FEMALE    HIGH SCHOOL 17551.25 MARRIED  40     65    156     37
4 16   MALE    HIGH SCHOOL 17551.25 MARRIED  56     NA    210      6
5 19   MALE LT HIGH SCHOOL 15703.75 MARRIED  51     68    170     NA
6 28   MALE       BACHELOR 73900.00 MARRIED  57     71    225     40
```

double filter

```
double_filtered <- gss %>%
    filter(marital == "MARRIED" | marital == "DIVORCED")
head(double_filtered)
```

```
  id    sex         degree   income  marital age height weight hrswrk
1  1   MALE       BACHELOR 60967.50 DIVORCED  53     72    190     60
2  2 FEMALE       BACHELOR 60967.50  MARRIED  26     60     97     40
3  4 FEMALE       BACHELOR 10161.25  MARRIED  56     68    160     20
4 14 FEMALE    HIGH SCHOOL 17551.25  MARRIED  40     65    156     37
5 16   MALE    HIGH SCHOOL 17551.25  MARRIED  56     NA    210      6
6 19   MALE LT HIGH SCHOOL 15703.75  MARRIED  51     68    170     NA
```

multiple filtering

```
gss_degree <- gss %>%
    filter(degree %in% c("BACHELOR", "GRADUATE", "JUNIOR COLLEGE"), marital %in%
        c("MARRIED", "DIVORCED"), sex %in% c("MALE"), )
head(gss_degree)
```

```
  id  sex         degree  income  marital age height weight hrswrk
1  1 MALE       BACHELOR 60967.5 DIVORCED  53     72    190     60
2 28 MALE       BACHELOR 73900.0  MARRIED  57     71    225     40
3 30 MALE       BACHELOR 33255.0 DIVORCED  54     71    165     40
4 40 MALE       BACHELOR 73900.0  MARRIED  35     NA    200     50
5 52 MALE       BACHELOR 40645.0  MARRIED  42     68    230     NA
6 53 MALE JUNIOR COLLEGE 40645.0  MARRIED  29     73    245     60
```

data format

long/wide format

sample data set

```
grades_wide <- tibble(student = c("alice", "Bob", "charlie"), math_2023 = c(85, 92,
    NA), math_2024 = c(88, 90, 85), science_2023 = c(78, NA, 80), science_2024 = c(82,
    85, 88))
head(grades_wide)
```

```
# A tibble: 3 × 5
  student math_2023 math_2024 science_2023 science_2024
  <chr>       <dbl>     <dbl>        <dbl>        <dbl>
1 alice          85        88           78           82
2 Bob            92        90           NA           85
3 charlie        NA        85           80           88
```

pivot_longer

- its purpose is to reduce the number of columns while keeping original info

```
grades_long <- grades_wide |>
    pivot_longer(cols = -student, names_to = c("subject", "year"), names_sep = "_",
        values_to = "score", )
head(grades_long)
```

```
# A tibble: 6 × 4
  student subject year  score
  <chr>   <chr>   <chr> <dbl>
1 alice   math    2023     85
2 alice   math    2024     88
3 alice   science 2023     78
4 alice   science 2024     82
5 Bob     math    2023     92
6 Bob     math    2024     90
```

pivot wider> long to wide format

```
grades_wide_back <- grades_long %>%
    pivot_wider(names_from = c("subject", "year"), names_sep = "_", values_from = score)
head(grades_wide_back)
```

```
# A tibble: 3 × 5
  student math_2023 math_2024 science_2023 science_2024
  <chr>       <dbl>     <dbl>        <dbl>        <dbl>
1 alice          85        88           78           82
2 Bob            92        90           NA           85
3 charlie        NA        85           80           88
```

advanced joints

```
df1 <- tibble(id = c(1, 2, 3, 4), name = c("alice", "Bob", "charlie", "David"))
df2 <- tibble(id = c(2, 3, 5), score = c(85, 90, 78))
```

join types

inner_join(): common matching keys, unique

```
inner_join(df1, df2, by = "id")
```

```
# A tibble: 2 × 3
     id name    score
  <dbl> <chr>   <dbl>
1     2 Bob        85
2     3 charlie    90
```

full_join()

```
full_join(df1, df2, by = "id")
```

```
# A tibble: 5 × 3
     id name    score
  <dbl> <chr>   <dbl>
1     1 alice      NA
2     2 Bob        85
3     3 charlie    90
4     4 David      NA
5     5 <NA>       78
```

semi-join

```
semi_join(df1, df2, by = "id")
```

```
# A tibble: 2 × 2
     id name
  <dbl> <chr>
1     2 Bob
2     3 charlie
```

```
semi_join(df2, df1, by = "id")
```

```
# A tibble: 2 × 2
     id score
  <dbl> <dbl>
1     2    85
2     3    90
```

anti_join()

```
anti_join(df1, df2, by = "id")
```

```
# A tibble: 2 × 2
     id name
  <dbl> <chr>
1     1 alice
2     4 David
```

left_join()

```
left_join(df1, df2, by = "id")
```

```
# A tibble: 4 × 3
     id name    score
  <dbl> <chr>   <dbl>
1     1 alice      NA
2     2 Bob        85
3     3 charlie    90
4     4 David      NA
```

right_join()

```
right_join(df1, df2, by = "id")
```

```
# A tibble: 3 × 3
     id name    score
  <dbl> <chr>   <dbl>
1     2 Bob        85
2     3 charlie    90
3     5 <NA>       78
```

join function using real data sets

load data sets

```
inv_1 <- read.csv("inv_1.csv")
inv_2 <- read.csv("inv_2.csv")
head(inv_1)
```

```
  Product_ID Purchase_Date    Category Region
1      P0415     5/28/2023    Clothing  South
2      P0463     6/26/2023    Clothing  South
3      P0179     5/31/2023       Books  North
4      P0526      5/3/2023    Clothing  North
5      P0195     6/24/2023 Electronics   East
6      P0938     8/16/2023       Books   East
```

```
head(inv_2)
```

```
  Product_ID Quantity Unit_Cost Selling_Price Discount Total_Cost Total_Revenue
1      P0415       62    249.80        456.94    31.51   15487.60      26376.66
2      P0463       69    377.40        986.74    20.63   26040.60      66661.59
3      P0179       30     88.69        150.71    18.27    2660.70       3973.20
4      P0526       62    422.60        835.50    31.04   26201.20      49876.52
5      P0195       45    163.55         39.23    47.80    7359.75       -385.65
6      P0938       68    120.77        780.97    23.80    8212.36      51487.56
    Profit Shipment_Date
1 10889.06     5/28/2023
2 40620.99     6/26/2023
3  1312.50     5/31/2023
4 23675.32      5/3/2023
5 -7745.40     6/24/2023
6 43275.20     8/16/2023
```

comined inventory by

```
combined_inv <- full_join(inv_1, inv_2, by = "Product_ID")
head(combined_inv)
```

```
  Product_ID Purchase_Date Category Region Quantity Unit_Cost Selling_Price
1      P0415     5/28/2023 Clothing  South       62    249.80        456.94
2      P0415     5/28/2023 Clothing  South       40    382.51        921.56
3      P0463     6/26/2023 Clothing  South       69    377.40        986.74
4      P0463     6/26/2023 Clothing  South        5    332.11        284.48
5      P0179     5/31/2023    Books  North       30     88.69        150.71
6      P0526      5/3/2023 Clothing  North       62    422.60        835.50
  Discount Total_Cost Total_Revenue   Profit Shipment_Date
1    31.51   15487.60      26376.66 10889.06     5/28/2023
2    24.35   15300.40      35888.40 20588.00     8/30/2023
3    20.63   26040.60      66661.59 40620.99     6/26/2023
4    48.69    1660.55       1178.95  -481.60     8/29/2023
5    18.27    2660.70       3973.20  1312.50     5/31/2023
6    31.04   26201.20      49876.52 23675.32      5/3/2023
```

transform data

```
combined_inv$Category <- as.factor(combined_inv$Category)
combined_inv$Region <- as.factor(combined_inv$Region)
head(combined_inv, 10)
```

```
   Product_ID Purchase_Date    Category Region Quantity Unit_Cost Selling_Price
1       P0415     5/28/2023    Clothing  South       62    249.80        456.94
2       P0415     5/28/2023    Clothing  South       40    382.51        921.56
3       P0463     6/26/2023    Clothing  South       69    377.40        986.74
4       P0463     6/26/2023    Clothing  South        5    332.11        284.48
5       P0179     5/31/2023       Books  North       30     88.69        150.71
6       P0526      5/3/2023    Clothing  North       62    422.60        835.50
7       P0195     6/24/2023 Electronics   East       45    163.55         39.23
8       P0195     6/24/2023 Electronics   East       57     19.14         25.93
9       P0938     8/16/2023       Books   East       68    120.77        780.97
10      P0938     8/16/2023       Books   East       83    201.08        438.08
   Discount Total_Cost Total_Revenue   Profit Shipment_Date
1     31.51   15487.60      26376.66 10889.06     5/28/2023
2     24.35   15300.40      35888.40 20588.00     8/30/2023
3     20.63   26040.60      66661.59 40620.99     6/26/2023
4     48.69    1660.55       1178.95  -481.60     8/29/2023
5     18.27    2660.70       3973.20  1312.50     5/31/2023
6     31.04   26201.20      49876.52 23675.32      5/3/2023
7     47.80    7359.75       -385.65 -7745.40     6/24/2023
8     30.61    1090.98       -266.76 -1357.74     3/23/2023
9     23.80    8212.36      51487.56 43275.20     8/16/2023
10    41.38   16689.64      32926.10 16236.46     2/11/2023
```